NeuroReg: Neural-inspired Adaptive Regulation for Dynamic Outlier Suppression in Fine-tuning

Anonymous Authors[1]

Abstract

Parameter-Efficient Fine-Tuning (PEFT) has become a popular approach for adapting large-scale language models (LLMs) to downstream tasks, enabling efficient updates by modifying only a small subset of parameters. However, PEFT faces persistent challenges, including training instability, slow convergence, and high performance variance. In this work, we identify and empirically characterize a critical yet overlooked failure mode in PEFT: dynamic anomaly propagation. Through systematic analysis, we show that it arises from two self-reinforcing loops: (1) Weight-Activation Coupling, where task-relevant weights are excessively amplified, causing activation outliers in specific dimensions, and (2) Representation-Gradient Degradation, where these activation outliers compress effective rank and destabilize gradients, exacerbating optimization challenges. To address this, we propose NeuroReg, a plug-and-play, real-time optimization-aware activation outlier suppression method inspired by the adaptive mechanisms of biological neurons. NeuroReg operates within each Transformer layer and incorporates dynamic activation outlier sensing, activation hard gating, and graded-response compression, adding less than 0.5% additional parameters. Extensive experiments on multiple LLMs and benchmarks demonstrate that NeuroReg significantly improves training stability, representation health, and downstream performance. Codes are available at `https://openreview.net/attachment?id=zTfR7OFUOK&name=supplementary_material`.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

In recent years, large-scale pre-trained language models (LLMs) such as GPT and LlaMA have advanced significantly via self-supervised learning (Hoffmann et al., 2022; Guo et al., 2025). Adapting these models to downstream tasks relies on fine-tuning—updating parameters using task-specific data (Dodge et al., 2020; Ding et al., 2023). Given the substantial size of modern LLMs, a widely used approach is Parameter-Efficient Fine-Tuning (PEFT), which updates only a small subset of parameters while keeping the majority fixed (Han et al., 2024; Xu et al., 2023). Despite its strong generalization potential, PEFT still faces notable challenges, including training instability, slow convergence, and high performance variance (Mosbach et al., 2020; Kumar et al., 2022).

Recent advances in PEFT can be organized into three complementary research directions that collectively address the training optimization challenges in adapting LLMs: convergence stability (Dodge et al., 2020; Hayou et al., 2024; Meng et al., 2024; Wang et al., 2024b), which improves optimization reliability through weight decomposition and refined initialization; dynamic structural adaptation (Zhang et al., 2023; Li et al., 2025; Valipour et al., 2023; He et al., 2025), which enhances parameter efficiency via adaptive low-rank modulation; and training process regulation (Jiang et al., 2025), which preserves representational fidelity through meta-supervised learning trajectories. While these methods have improved fine-tuning performance, they rely on a key assumption: that pre-trained weights are well-conditioned, providing a stable starting point for optimization. Based on this assumption, they typically explore a parameter subspace near the pre-trained weights to find a local optimum that minimizes downstream task loss. However, we challenge this assumption. Recent studies suggest that pre-trained weights exhibit structural flaws (Lee et al., 2024; Dettmers et al., 2024; Lin et al., 2024). Specifically, task-critical features become concentrated in only a few dimensions. The connection weights in these dimensions are amplified to magnitudes higher than others, which are known as anomalous weights. Furthermore, the activation of these anomalous weights by task-specific inputs leads to activation outliers. Such structural issues disrupt the processes of feature repre-

sentation and gradient optimization, limiting the practical effectiveness of current fine-tuning methods.

Notably, existing studies have explored various approaches to handling model activation outliers at different stages. They can be categorized into two main approaches: static adjustment during post-processing (Xiao et al., 2023; Lin et al., 2024; Dettmers et al., 2022) and indirect stabilization during pre-training (Shen et al., 2020; Lubana et al., 2021; Wang et al., 2024a). The former relies on predefined rules and focuses on static quantization operations, making it difficult to adapt to the dynamic changes in activation distributions during downstream tasks. The latter, while addressing training dynamics, is limited to adjustments at the distributional level and cannot directly intervene in the transient outlier activation spikes that occur during forward propagation. The shared limitation of these two approaches lies in their inability to dynamically respond to the real-time, evolving patterns of activation outliers in fine-tuning scenarios. Furthermore, they fail to integrate activation outliers as structural defects in training with optimization dynamics, making them ineffective in addressing the deeper impacts of outliers on the training process.

In this work, we identify a critical yet overlooked failure mode in LLM fine-tuning: dynamic anomaly propagation. Through systematic analysis, we reveal that it stems from two self-reinforcing loops: 1) Weight-Activation Coupling: In the early stage of fine-tuning, task-relevant weights are excessively amplified, triggering activation outliers in specific dimensions. 2) Representation-Gradient Degradation: These activation outliers compress the effective rank in the forward pass while destabilizing gradients in the backward pass, further aggravating weight anomalies and deteriorating the optimization trajectory. To break this loop, we introduce NeuroReg, a plug-and-play, real-time optimization-aware activation outlier suppression method inspired by the dynamic adaptive mechanism of biological neurons. It is embedded into each Transformer layer and features three designs: dynamic activation outlier sensing, activation hard gating, and graded-response compression. With under 0.5% added parameters, our plug-and-play module enhances training stability and boosts downstream performance across tasks. Our main contributions can be summarized as follows:

- We identify and empirically characterize dynamic anomaly propagation in PEFT, showing it arises from two coupled loops: weight-activation coupling and subsequent representation-gradient degradation.
- We propose NeuroReg, a plug-and-play, real-time optimization-aware activation outlier suppression method in each Transformer layer. It integrates dynamic sensing, activation hard gating, and graded compression adding under 0.5% parameters.
- We validate NeuroReg on multiple LLMs across diverse benchmarks, demonstrating consistent improvements in representation health, training stability, and downstream performance.

2. Related Work

Efficient Fine-Tuning Paradigms for LLMs. To improve training optimization, efficient fine-tuning methods for large language models (LLMs) primarily revolve around three key research directions: convergence stability, dynamic structural adaptation, and training process regulation. Convergence stability approaches, such as DoRA(Liu et al., 2024) and LoRA+(Hayou et al., 2024), enhance training robustness through weight decomposition and hierarchical learning rate scheduling. PISSA(Meng et al., 2024) and LoRA-GA(Wang et al., 2024b) further improve initialization quality and gradient alignment. In the domain of dynamic structural adaptation, methods like AdaLoRA(Zhang et al., 2023), TopLoRA(Li et al., 2025), and DyLoRA(Valipour et al., 2023) boost parameter efficiency by dynamically adjusting low-rank dimensions or enabling adaptive rank configuration during training. On the other hand, GoRA(He et al., 2025) enhances model expressiveness through gated orthogonal transformations. However, the adaptive capabilities of these methods overlook the structural defects introduced by pre-trained anomalous weights.

Activation Outlier Suppression in Deep Models. Existing methods for modulating activation outliers can be broadly categorized into two classes. The first class comprises deployment-oriented static numerical reformation techniques, such as SmoothQuant(Xiao et al., 2023), LLM-int8(Dettmers et al., 2022), and ZeroQuant(Yao et al., 2022). These methods calibrate the outlier distribution in post-trained models using offline statistics to improve quantization stability. The second class consists of training-oriented dynamic distribution regulation approaches, including RMSNorm(Zhang & Sennrich, 2019), PowerNorm(Shen et al., 2020), and DeepNorm(Wang et al., 2024a). These methods adjust activation distributions globally or at the module level during training. Nevertheless, they struggle to precisely identify and suppress the sparse, transient outlier peaks that occur in the forward pass, which are the direct cause of gradient and representation degradation.

Neural-inspired Adaptive Computation. Principles from biological neural systems have inspired robust and efficient algorithms, particularly in processing time-varying signals (Gast et al., 2024; Zheng et al., 2024). The widely used Leaky Integrate-and-Fire (LIF) neuron in Spiking Neural Networks (SNNs) (Ghosh-Dastidar & Adeli, 2009; Gerstner et al., 2014), for instance, employs membrane-potential-like integration, threshold-driven firing, and adaptive reset mechanisms to handle dynamic inputs with high energy efficiency.

Recent work has explored importing such concepts into deep learning, especially in tasks with complex dynamic structures (Liao et al., 2024; Yan et al., 2024; Zhang et al., 2024).However, there is a lack of addressing training-time instability arising from anomalous activation dynamics.

3. A Heuristic Analysis of Dynamic Anomaly Propagation in PEFT

This section provides a heuristic analysis of the dynamic anomaly propagation in fine-tuning: 1) Weight-Activation Coupling; 2) Representation-Gradient Degradation, creating a self-reinforcing unstable training loop. Existing methods, relying on well-behaved weight assumptions or static and globally uniform regulation, fail to intervene in this dynamic process, thereby motivating the design goals of our method. This analysis is grounded in quantifiable empirical observations from LLMs (see Appendix A). While the construction of the dynamic framework is heuristic, the subsequent derivations—specifically the eigenvalue dominance affecting the effective rank and the gradient scaling with input magnitude—are rigorously conducted using linear algebra and calculus.

3.1. Weight-Activation Coupling

Formalization of Fine-tuning. We define the pre-trained model as a parametric function $F(\mathbf{X};\mathbf{\Theta}_0)$ with L layers, where $\mathbf{X}$ denotes the input data from the downstream task domain, and $\mathbf{\Theta}_0$ consists of the layer-wise pre-trained parameters $\mathbf{W}^\ell$, structured as $\mathbf{\Theta}_0 = \left\{\mathbf{W}^\ell\right\}_{\ell=1}^{L}$. Fine-tuning seeks a parameter increment $\mathbf{\Delta\Theta}$ to minimize task loss $\mathcal{L}$ on a new distribution $\mathcal{D}_{\text{new}}$ of the downstream task. We focus on the Transformer's core operation in layer l, which projects the input $\mathbf{h}^{l-1}$ via:

$$\mathbf{z}_P^l = (\mathbf{W}_0^l + \Delta\mathbf{W}^l)\mathbf{h}^{l-1}. \tag{1}$$

Here, $\mathbf{W}^\ell$ denotes the complete set of trainable parameters in layer ℓ, encompassing the attention weight matrices ($\mathbf{W}_q^\ell$, $\mathbf{W}_k^\ell$, $\mathbf{W}_v^\ell$), the feed-forward weight matrix $\mathbf{W}_{\text{FFN}}^\ell$, and the layer-normalization parameters ($\boldsymbol{\gamma}^\ell$, $\boldsymbol{\beta}^\ell$). $\Delta\mathbf{W}^l$ is the fine-tuning increment.

Anomalous Weights. Existing studies have shown that pre-trained weight matrices contain prominent numerical outliers: a small subset of dimensions exhibit parameter magnitudes that exceed the typical range by orders of magnitude, dominating the model's forward computation and gradient dynamics (Lee et al., 2024; Yuan et al.; Dettmers et al., 2024). Motivated by this observation, we decompose the weight matrix of each layer into **normal** and **anomalous** components:

$$\mathbf{W}_0^l = \mathbf{W}_{\text{normal}}^l + \mathbf{W}_{\text{anomalous}}^l \tag{2}$$

Here, $\mathbf{W}\text{normal}^l$ and $\mathbf{W}\text{anomalous}^l$ correspond to the bulk weights and the structured outlier weights, respectively. Although anomalous weights constitute an extremely small proportion in quantity, they dominate the model's expressive power and often induce gradient instability and training oscillations during fine-tuning.

Activation Outliers. Recent studies(Lee et al., 2024; Dettmers et al., 2024; Lin et al., 2024) have demonstrated that pre-trained anomalous weights, when activated by task-specific inputs during fine-tuning, serve as the primary origin of activation outliers. This behavior is characterized by a disproportionate amplification of responses along particular feature directions. We formalize this task-conditioned amplification property of anomalous weights as follows:

Definition 3.1 (Activation Outliers)**.** Let $\mathbf{z}^l = \mathbf{W}^l\mathbf{h}^{l-1}$ be the linear projection output at layer l. If there exists an anomalous weight direction $\mathbf{v}_*$ (satisfying Definition 3.1) and a downstream input $\mathbf{h}^{l-1}$ such that the projection of $\mathbf{h}^{l-1}$ onto this direction satisfies $|\mathbf{h}^{l-1}\cdot\mathbf{v}_*| > 0$, and the corresponding activation output fulfills

$$\|\mathbf{z}^l\| \geq \kappa \cdot \mathbb{E}_{\mathcal{D}_{\text{new}}}\left[\|\mathbf{z}^l\|\right], \quad \kappa \gg 1, \tag{3}$$

where κ is the anomaly-activation threshold, then $\mathbf{z}^l$ is called an activation outlier and denoted as $\mathbf{z}_{\text{outlier}}^l$.

Remark. To understand the root causes of instability in LLM fine-tuning and to provide guidance for method design, we present the following analysis. This analysis is grounded in empirical observations of LLM activation patterns sand aims to offer a plausible causal explanation along with a heuristic framework.

3.2. Representation-Gradient Degradation

Forward: Activation Outlier to Rank Collapse. The activation outliers dominate the statistics of the feature matrix $\mathbf{Z}_P^l \in \mathbb{R}^{n\times d}$ (where n is the sequence length and d is the feature dimension). According to random matrix theory(Pennington & Worah, 2017), when the covariance of a data matrix is dominated by a small number of strong components, its eigenvalue distribution becomes highly skewed. This dominance by a few oversized eigenvalues mathematically corresponds to a collapse in the effective rank of the representation—a phenomenon that has been empirically observed in large-scale Transformers as "representation degeneration" or "rank collapse" (Huang et al., 2020; You et al., 2022)(see Appendix A). We illustrate it as follows:

Heuristic Corollary 3.2 (Activation Outlier-Induced Effective Rank Collapse) Let $\mathbf{Z}_P^l$ be the feature matrix at layer l, and let $\Sigma^l = \mathrm{Cov}(\mathbf{Z}_P^l)$ be its covariance matrix. Under Definition 3.1 and assuming the anomalous component's activation is sufficiently dominant, the **effective (stable) rank** of $\mathbf{Z}_P^l$ is upper-bounded by a constant independent of d:

$$R_{\text{eff}}(\mathbf{Z}_P^l) := \frac{\left(\sum_{i=1}^{d} \sigma_i(\Sigma^l)\right)^2}{\sum_{i=1}^{d} \sigma_i^2(\Sigma^l)} \leq C \ll d \tag{4}$$

Proof Sketch. The condition in Definition 3.1 implies that the covariance matrix Σ^l has a leading singular value σ_1 that dominates the rest, i.e., $\sigma_1 \geq \kappa \sum_{i>1} \sigma_i$ for some $\kappa > 1$. Substituting this inequality into the definition of R_{eff} yields the constant upper bound $C \approx 1 + 1/\kappa$. The full proof is provided in Appendix A. □

Backward: Gradient and Anomaly Dynamics. The instability stems from the high sensitivity of the gradient $\mathbf{g}_P^l$ to activation outliers $\mathbf{z}_P^l$. According to chain-rule analysis and Lipschitz continuity, an abnormally large magnitude of the input activation directly and proportionally amplifies the gradient norm (Glorot & Bengio, 2010). Therefore, we explicitly trace the root of this optimization instability back to the structural weight anomalies in the pre-trained model (Equation 2) and the dynamic activation outliers they induce during forward propagation (Definition 3.1). We now formalize this feedback effect:

$$\frac{\partial \mathcal{L}}{\partial \Delta \mathbf{W}^l} = \mathbf{g}_P^l (\mathbf{h}^{l-1})^\top, \quad \text{where} \quad \mathbf{g}_P^l = \frac{\partial \mathcal{L}}{\partial \mathbf{z}_P^l} \tag{5}$$

Heuristic Corollary 3.2 (Gradient Amplification of Anomalies) Consider the gradient update for the weight increment. Under standard Lipschitz and smoothness assumptions (a standard assumption in optimization theory for analyzing gradient dynamics(Bottou et al., 2018) and is commonly adopted in fine-tuning studies(Hayou et al., 2024)) of the loss, if the forward pass produces an activation outlier satisfying $\left\|\mathbf{z}_{P,\text{outlier}}^l\right\| \geq \xi$, then the component of the gradient update aligned with the anomalous weight direction $\mathbf{W}_{\text{anomalous}}^l$ scales at least linearly with ξ:

$$\left\|\mathrm{Proj}_{\mathbf{W}_{\text{anomalous}}^l}\left(\frac{\partial \mathcal{L}}{\partial \Delta \mathbf{W}^l}\right)\right\|_F \geq \alpha \cdot \xi - \beta \tag{6}$$

where $\alpha, \beta > 0$ are constants depending on the task loss and network architecture. Consequently, the anomalous weight component receives disproportionately large updates, reinforcing its magnitude in subsequent steps.

Proof Sketch. The chain rule links $\mathbf{g}_P^l$ to $\mathbf{z}_P^l$ via the network's Jacobian. The dominance of $\mathbf{z}_{P,\text{outlier}}^l$ forces a large component of $\mathbf{g}_P^l$ to align with the sensitive direction of the loss. Projecting this gradient onto the subspace of $\mathbf{W}_{\text{anomalous}}^l$ yields the lower bound (6). The full proof is provided in Appendix B. □

3.3. Dynamic Anomaly Propagation

Heuristic Corollary 3.2 and 3.3 define a discrete dynamical system. This system captures a positive feedback loop where activation outliers distort gradients, which in turn amplify the underlying anomalies. Specifically, the cycle proceeds as follows: **Activation:** downstream data activate $\mathbf{W}_{\text{anomalous}}^l$ via Eq.(2) → **Rank Collapse:** causes forward-feature rank collapse (Heuristic Corollary 3.2) → **Gradient Oscillation:** activation outliers induce large, biased gradients (Heuristic Corollary 3.3) → **Amplification:** gradients amplify $\mathbf{W}_{\text{anomalous}}^l$, reinforcing the cycle in the next iteration.

Formally, the feedback loop can be expressed as:

$$\begin{cases} \mathbf{z}_{P,(t)}^l = \left(\mathbf{W}_{\text{abnormal}}^l + \Delta \mathbf{W}_{(t-1)}^l\right) \mathbf{h}_{(t)}^{l-1} \\ \Delta \mathbf{W}_{(t)}^l = \Delta \mathbf{W}_{(t-1)}^l - \eta \cdot \mathcal{O}\left(\mathbf{z}_{P,(t)}^l, \mathbf{h}_{(t)}^{l-1}\right) \end{cases} \tag{8}$$

where $\mathcal{O}(\cdot)$ represents the optimizer's update rule. The system's sensitivity to $\mathbf{z}_{P,(t)}^l$ creates a self-reinforcing feedback loop that persistently destabilizes training.

The dynamical system in Equation (8) is a heuristic simplification of the actual training dynamics. It is intended to illustrate the positive feedback loop conceptually.

Why Existing Methods Fail. This analysis clarifies the limitations: 1) Fine-tuning methods assume $\mathbf{W}_0^l$ is well-conditioned. They cannot break the cycle because they do not detect or regulate the dynamic activation outliers. 2) Static outlier suppression uses fixed thresholds, which are inherently insensitive to structural evolution. They therefore fail to adapt to the time-varying outlier patterns implied by Heuristic Corollary 3.2 and the continuously reorganized projection geometry captured by the evolving dynamics of Equation 8. 3) Normalization methods during training operate within the training dynamics, but their adjustment mechanisms focus primarily on the distributional level (e.g., covariance stabilization, mean-variance normalization). They fail to directly and precisely suppress the sparse, instantaneous activation peaks that arise during forward propagation. These peaks are the direct cause of gradient oscillations and representational rank collapse.

Conclusion. The dynamic anomaly propagation is a self-reinforcing structural dynamic defect, not a static one. Effective intervention must therefore be inherently dynamic: it must 1) operate in real time to intercept each forward pass, 2) target the activation stream at its origin before it corrupts gradients, and 3) continuously adapt its suppression policy

to the evolving activation outlier distribution. This diagnosis directly motivates the design of our neural-inspired adaptive regulation paradigm.

4. Method

4.1. Overview of NeuroReg

Based on the analysis in Section 3, we identify three design guidelines that our method must follow: real-time intervention, activation-stream targeting, and continuous adaptation. Notably, these principles align closely with the dynamic modulation mechanisms of the LIF neuron in SNNs–specifically its threshold-driven firing behavior, sparse and efficient spike propagation, and adaptive response decay. Inspired by this, we propose NeuroReg, a lightweight, plug-and-play adaptive regulation framework that draws on neuronal information-processing principles. In implementation, we embed a lightweight neural regulation unit into each Transformer layer. This unit sequentially performs three computational steps that correspond to neuronal mechanisms: 1) Dynamic Activation Outlier Sensing. 2) Activation Hard Gating. 3) Graded-Response Compression.

4.2. Dynamic Activation Outlier Sensing

To fulfill the requirement of continuous adaptation (Principle 3), the NeuroReg module first detects activation outliers in real-time. The detection threshold T is computed per layer using a dual-benchmark strategy that adapts to the evolving activation statistics:

$$T = \max\Big(\underbrace{P_{90}(\|\mathbf{H}\|)}_{\text{adaptive benchmark}} , \underbrace{3 \times \text{median}(\|\mathbf{H}\|)}_{\text{absolute benchmark}}, \underbrace{1.0}_{\text{numerical stability}} \Big) \tag{7}$$

The adaptive benchmark (90th percentile) identifies relatively extreme values within the current batch, automatically raising the threshold when outliers are prevalent to avoid over-suppression. Concurrently, the absolute benchmark (three times the median) safeguards sensitivity to isolated small-magnitude anomalies. This design ensures the threshold evolves with the time-varying outlier distribution .

4.3. Activation Hard Gating

To precisely isolate activation outliers immediately after detection, inspired by the "0-1" firing principle of LIF neurons, NeuroReg employs a deterministic gating mechanism.

Based on the adaptive threshold T, a binary hard mask $\mathbf{M}$ is generated for the current activation tensor $\mathbf{H}$:

$$eq8\mathbf{M}_{b,t,i} = \mathbb{I}(\|\mathbf{h}_{b,t,i}\| > T) \in \{0, 1\}, \tag{8}$$

where b, t, i index the batch, token position, and feature dimension, respectively.

This gating strategy embodies three key properties that align with our design imperatives: 1) Deterministic & Lightweight: The decision is based solely on the computed statistical threshold T, introducing no randomness and negligible computational overhead. 2) Sparse & Precise: The mask $\mathbf{M}$ is concentrated in a subset of tokens, which will be gradually sparse during training. This ensures that computational resources are focused precisely on the targeted anomalous regions without disturbing the vast majority of normal signals. 3) Gradient-Flow Preserving: The mask $\mathbf{M}$ is detached from the computational graph. During backpropagation, $\mathbf{M}$ is a constant coefficient that selects gradient paths: anomalous positions receive gradients transformed by f_{compress}, while normal positions retain their original gradients.

4.4. Graded-response Compression

Inspired by the adaptive decay mechanism of biological neurons, this stage performs adaptive suppression while preserving useful information, allowing the model to dynamically adjust its response based on signal intensity.

4.4.1. Learnable Compression for Outliers

For activations $\mathbf{z}$ flagged as abnormal ($\mathbf{M} = 1$), NeuroReg applies a dimension-wise, learnable linear compression:

$$\mathbf{c}_{\text{abn}} = \sigma(\boldsymbol{\theta}_s) \odot \mathbf{z}, \tag{9}$$

where $\boldsymbol{\theta}_s \in \mathbb{R}^d$ is a learnable sensitivity parameter vector (one per feature dimension), and $\sigma(\cdot)$ denotes the sigmoid function, constraining the compression coefficient to the interval $(0, 1)$.

The compression rate (CR) for each dimension is defined as the ratio of output to input magnitude: $\text{CR} = \sigma(\boldsymbol{\theta}_s)$. Notably, these rates are not pre-defined hyperparameters but are optimized end-to-end via gradient descent. This allows NeuroReg to automatically and adaptively discover the optimal suppression intensity for the anomalous component of each feature dimension throughout training.

4.4.2. Identity Pass-Through and Gated Fusion

For activations in the normal region ($\mathbf{M} = 0$), NeuroReg employs an identity pass-through $\mathbf{c}_{\text{normal}} = \mathbf{z}$ to ensure no distortion. This guarantees that task-relevant information carried by the majority of normal activations remains entirely intact, preserving the model's foundational capacity.

Finally, the regulated activation is obtained through an element-wise gated fusion of the two processed streams:

$$\mathbf{c} = \mathbf{m} \odot \mathbf{c}_{\text{outlier}} + (1 - \mathbf{m}) \odot \mathbf{c}_{\text{normal}}. \tag{10}$$

This operation reintegrates the adaptively compressed anomalies with the pristine normal signals, producing a

stabilized activation output that is ready for subsequent layers in the forward pass.

5. Experiments

To comprehensively evaluate NeuroReg, we compare it with baseline methods across three tasks: natural language understanding (fine-tuning LlaMA2-7B/LlaMA3-8B on commonsense reasoning), mathematical reasoning (fine-tuning Mistral-7B/Gemma-2 9B on MetaMathQA; evaluated on GSM8K/MATH), and image classification (fine-tuning CLIP-ViT-B/16 on seven datasets). Results and analysis of image classification tasks can be found in Appendix E.2. Additionally, a detailed computation cost analysis of our method is provided in Appendix C. NeuroReg improves the average accuracy of all evaluated PEFT methods across three tasks (language understanding, mathematical reasoning, and image classification), with gains ranging from 0.3% to 3.1%, while enhancing training stability and representation health.

5.1. Evaluation on Language Understanding Tasks

We fine-tune Llama2-7B and Llama3-8B on a collection of commonsense reasoning tasks comprising eight sub-tasks, each with its own predefined training and test splits. More Experimental details are provided in Appendix C. The results for Llama2-7B are summarized in Table 1 (results for Llama3-8B are provided in Appendix E.1)). NeuroReg consistently improves the performance of every PEFT method it is paired with. When added to standard LoRA (Hu et al., 2022), NeuroReg+LoRA achieves an average accuracy of 78.2%, surpassing vanilla LoRA 77.6% by 0.6%. More notably, NeuroReg substantially lifts weaker baselines: LoRA+ 74.1% is improved to NeuroReg+LoRA+ 74.6%, a gain of 0.5%; AdaLoRA 70.7% rises to NeuroReg+AdaLoRA 71.7%, a 1.0% increase; and the strongest baseline, LoRA-SB(Ponkshe et al., 2024) 81.2%, is further boosted by NeuroReg to 81.5%. These improvements are observed consistently across tasks, especially on complex benchmarks such as HellaSwag and ARC-Challenge, demonstrating that NeuroReg's dual mechanism of activation outlier suppression and gradient balancing effectively stabilizes fine-tuning. The results also highlight that NeuroReg's plug-in characteristic: it can integrate with diverse PEFT strategies (standard LoRA, LoRA+, LoRA-SB, AdaLoRA, etc.) without requiring method-specific adjustments.

5.2. Evaluation on Mathematical Reasoning Tasks

We fine-tune Mistral-7B(Jiang et al.) and Gemma-2 9B(Team et al., 2024) models on a subset of 50K samples from the MetaMathQA(Yu et al., 2023) dataset and evaluate performance on the GSM8K(Cobbe et al., 2021) and MATH(Hendrycks et al., 2021) mathematical reasoning benchmarks. More Experimental details are provided in Appendix C. The results are summarized in Table 2. Results demonstrate that all NeuroReg-enhanced variants consistently outperform their corresponding baselines. Notably, NeuroReg+LoRA-Pro(Wang et al., 2024b) achieves the highest performance across both tasks and models, scoring 63.7 on GSM8K and 17.9 on MATH for Mistral-7B, and 79.6 on GSM8K and 37.4 on MATH for Gemma-2 9B. This underscores the efficacy of integrating structured regularization with advanced low-rank adaptation. Particularly noteworthy is that NeuroReg+DoRA attains a score of 37.8 on the MATH benchmark for Gemma-2 9B, surpassing all other methods on this task. This suggests that NeuroReg's generalization-enhancing mechanism complements DoRA's weight-normalized decomposition effectively. Moreover, even the relatively weaker baseline method LoRA-SB shows meaningful improvement when augmented with NeuroReg, that is, gaining 1.6 points on GSM8K and 0.7 points on MATH for Mistral-7B while maintaining comparable parameter efficiency. This indicates that NeuroReg not only enhances high-performing methods but also provides consistent benefits to lightweight adapters.

5.3. Training Dynamics Analysis

To investigate NeuroReg's optimization mechanism during training, we select the widely used LoRA method along with three representative SOTA baselines: convergence stabilization methods (LoRA-SB), dynamic structural adaptation methods (TopLoRA), and training process regulation methods (Norm). By comparing the training dynamics of these baseline methods and their "baseline + NeuroReg" variants on the LlaMA2-7B model for commonsense reasoning tasks, we analyze micro-level dimensions such as the health of forward representations and the stability of backward gradients, ultimately assessing macro-level optimization effects through loss curves.

Forward Representation Health. To compare forward representation health during fine-tuning, we analyze LoRA and NeuroReg+LoRA (other methods are in Appendix F). We track the co-evolution of activation outlier (measured by kurtosis (He et al., 2024)) and effective rank across selected layers (5, 14, 23, 32). As shown in Figure 1(a), baseline LoRA exhibits rising kurtosis coupled with declining effective rank, reflecting typical overfitting where activations sharpen while representational capacity degrades. In contrast, NeuroReg+LoRA (Figure 1(b)) reverses this trend: activation outliers are suppressed, and effective rank increases and stabilizes in a higher range (100–200). This demonstrates NeuroReg's dual regularization mechanism: it dynamically suppresses activation outliers to mitigate distribution shift, while preserving higher effective rank to maintain representation diversity. Thus, it prevents the cascade collapse often seen in conventional PEFT—"activation

Table 1. Performance comparison of LlaMA2-7B with various PEFT methods on eight commonsense reasoning datasets.

PEFT Method	BoolQ	PIQA	SIQA	HellaSwag	WinoGrande	ARC-e	ARC-c	OBQA	Avg.
LoRA	69.8	79.9	79.5	83.6	82.6	79.8	64.7	81.0	77.6
NeuroReg+LoRA	70.2	80.4	79.8	85.4	83.2	80.2	65.2	81.4	78.2
Convergence Stability Methods									
PiSSA	67.6	78.1	78.4	76.6	78.0	75.8	60.2	75.6	73.8
LoRA+	69.4	80.1	80.6	73.4	78.3	73.7	59.3	78.2	74.1
DoRA	72.0	83.1	79.9	89.1	83.0	84.5	71.0	81.2	80.5
LoRA-SB	71.3	**85.1**	80.8	90.4	83.7	84.6	70.6	**82.8**	81.2
NeuroReg+LoRA+	70.3	80.5	80.3	73.8	78.6	74.2	60.1	79.0	74.6
NeuroReg+DoRA	**72.8**	82.8	80.3	90.5	83.7	82.8	68.3	82.4	80.5
NeuroReg+LoRA-SB	72.4	84.9	**81.5**	**92.4**	84.2	84.2	69.6	**82.8**	**81.5**
Dynamic Structural Adaptation Methods									
AdaLoRA	62.1	83.2	66.4	42.1	78.6	84.1	69.5	79.2	70.7
TopLoRA	70.5	83.1	81.0	87.2	83.6	83.9	68.2	82.2	80.0
NeuroReg+AdaLoRA	62.5	83.9	67.0	47.1	78.4	**84.5**	70.1	80.0	71.7
NeuroReg+TopLoRA	70.8	83.6	80.8	87.3	84.3	84.2	69.0	82.4	80.3
Training Process Regulation Methods									
Norm	69.7	83.4	78.6	87.2	81.0	81.9	66.2	79.2	78.4
NeuroReg+Norm	72.5	84.9	79.8	92.1	**84.7**	84.2	**71.2**	82.4	**81.5**

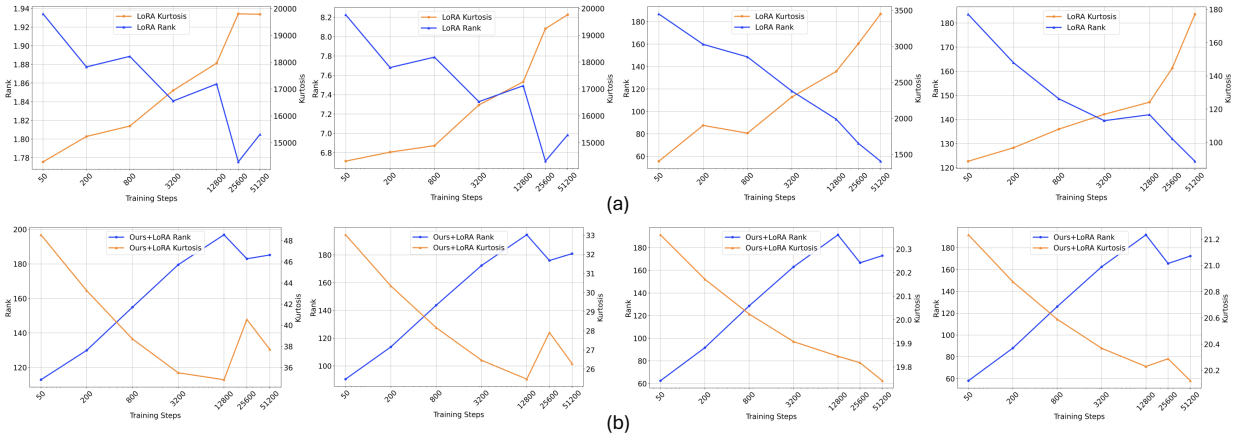

Figure 1. Evolution of activation kurtosis and effective rank in LlaMA2-7B layers 5, 14, 23, and 32 during fine-tuning with LoRA (a) and NeuroReg+LoRA (b).

Table 2. Performance comparison of PEFT methods on Mistral-7B and Gemma-2 9B across the arithmetic reasoning benchmarks GSM8K and MATH.

Method	Mistral-7B		Gemma-2 9B	
	GSM8K	MATH	GSM8K	MATH
LoRA	61.9	16.0	76.2	36.6
PiSSA	62.4	16.5	77.1	37.0
DoRA	62.7	16.6	77.6	37.0
LoRA-Pro	63.1	17.3	78.3	37.5
LoRA-SB	58.9	15.3	75.4	36.7
NeuroReg+LoRA	62.8	17.2	78.0	37.2
NeuroReg+DoRA	63.4	17.3	79.2	**37.8**
NeuroReg+LoRA-Pro	**63.7**	**17.9**	**79.6**	37.4
NeuroReg+LoRA-SB	60.5	16.0	76.8	36.5

outlier accumulation → effective rank decay → representation degradation."

Backward Gradient Stability. Figure 2 illustrates the evolution of gradient norms across layers during fine-tuning with LoRA and NeuroReg+LoRA (other methods in Appendix F). Baseline LoRA shows pronounced gradient concentration in middle layers, with norms up to $4.5\times$ above the layer-wise mean, indicating imbalanced optimization. NeuroReg consistently moderates this imbalance: 1) Early (3,200 steps): enhances gradients in under-activated layers for better anomaly modulation; 2) Mid-phase (12,800 steps): achieves more uniform gradient distribution while matching overall magnitude; 3) Late (25,600–51,200 steps): reduces total gradient magnitude for stable convergence while preserving balanced inter-layer flow. These results show that NeuroReg balances gradient propagation, mitigates abnormal aggregation, and promotes more stable and uniform optimization.

Loss Convergence Smoothness. Figure 3 compares the loss curves of LoRA and NeuroReg+LoRA during training (results for other methods are provided in Appendix F). The results show that NeuroReg+LoRA converges slightly

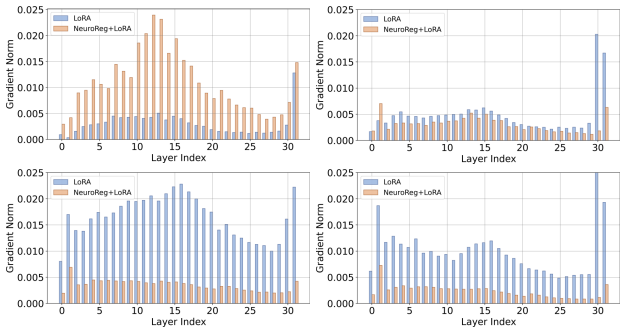

Figure 2. Comparison of gradient norms in different layers at training steps 3,200, 12,800, 25,600, and 51,200 for LoRA and NeuroReg+LoRA.

slower than the LoRA baseline in the early phase, reflecting its active anomaly modulation learning. As training proceeds, NeuroReg+LoRA not only reaches a more stable loss plateau but also converges to a slightly lower final loss value. These dynamics indicate that NeuroReg synergistically enhances training stability through a dual mechanism: suppressing abnormal forward activations and balancing backward gradient distributions across layers.

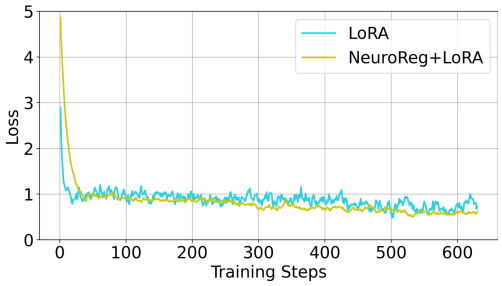

Figure 3. Training loss of LoRA and NeuroReg+LoRA.

5.4. Component Ablation Study

To validate the necessity of each core component in NeuroReg and examine its dynamic effects during training, we use NeuroReg+LoRA as the baseline and conduct a comprehensive ablation study on LlaMA2-7B using commonsense reasoning tasks. Table A.3 reports the final task accuracy and the dynamic evolution of training health metrics under different ablation settings (Figs. 4–6).

Adaptive Dynamic Thresholding via Layer-wise Statistics. Table A.3 shows that replacing the adaptive layer-wise threshold (Eq. 9) with a data-agnostic global learnable threshold (NeuroReg + LoRA*) reduces average accuracy by 1.5%. Fig. 4 compares how NeuroReg's threshold and a global threshold track the 90th percentile of activations across layers 1, 16, and 32. NeuroReg's threshold closely follows the dynamic activation evolution per layer, while the global threshold decouples from actual distributions. This failure to modulate outliers leads the model to optimize along pathological directions, causing persistently high activations and continuous effective-rank decay. In contrast, full NeuroReg maintains stable, effective rank and effective outlier control.

Figure 4. Comparison between NeuroReg's adaptive threshold and a globally learnable threshold in tracking the 90th percentile of activations throughout training, shown for layers 1, 16, and 32.

Deterministic Hard Sparse Gating. Replacing the hard binary gating in Eq. 10 with a soft sigmoid gate ($\mathbf{M}_{\text{soft}} = \text{sigmoid}\big((|\mathbf{h}| - T)/\tau\big)$, $\tau = 0.5$, denoted NeuroReg + LoRA$^+$) introduces indiscriminate activation modulation and noisy gradient updates. As shown in Table A.3, this reduces final accuracy by 0.2% and disrupts training health. Fig. 5 compares gradient-norm distributions across layers at steps 3,200, 12,800, 25,600, and 51,200. Hard gating maintains a stable, compact inter-layer gradient-norm distribution across checkpoints, preserving gradient purity and avoiding outlier contamination. In contrast, soft gating exhibits higher volatility and dispersion in gradient norms, with a significantly larger inter-layer coefficient of variation. These results confirm that deterministic, sparse isolation of activation outliers is essential for gradient stability and balanced layer-wise optimization.

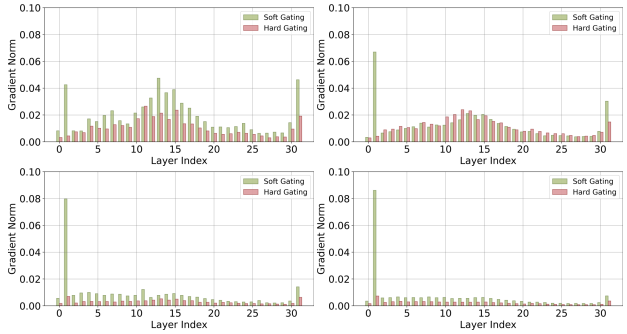

Figure 5. Comparison of gradient norm distributions across layers at training steps 3,200, 12,800, 25,600, and 51,200 under hard and soft gating.

Learnable Dimensional Compression. We compare NeuroReg against its variant with a globally shared compression parameter (NeuroReg + LoRA$^\diamond$) by analyzing the top-5% outliers in the most affected layer (layer 15) and their assigned compression strengths at three training stages (3,200, 25,600, and 51,200 steps) in Fig. 6. Replacing the learnable per-dimension compression (Eq. 9) with a shared global parameter leads to uniform, undifferentiated activation modulation. Fig. 6 shows that the shared parameter applies a "one-size-fits-all" scaling to the top-5% outliers in layer 15 across all stages, risking suppression of normal activations. In contrast, NeuroReg adaptively allocates compres-

sion strength based on the data distribution and training phase, enabling precise outlier-aware modulation. This adaptability yields more stable optimization, whereas fixed compression lowers final accuracy 77.8%. These results confirm that tailored, dimension-wise compression is essential for performance and training stability.

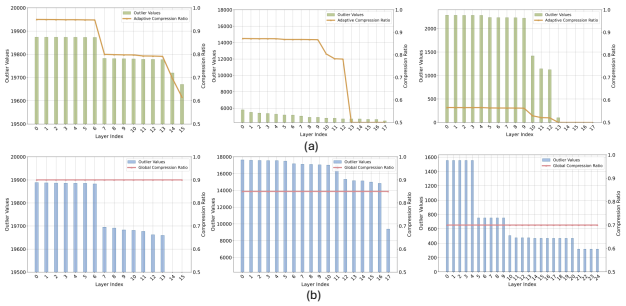

Figure 6. Comparison of top-5% outliers and their corresponding compression strengths under NeuroReg (a) and a globally-shared compression parameter (b) at three training stages (3,200, 25,600, and 51,200 steps).

6. Conclusion

This work identifies dynamic anomaly propagation in LLM fine-tuning—where activation outliers degrade both representation and gradient flow, creating a self-reinforcing failure cycle. We introduce NeuroReg, a lightweight plug-and-play module that dynamically suppresses outliers while preserving normal signals via ¡0.5% extra parameters. NeuroReg consistently improves representation health, training stability, and task performance, offering a general solution for efficient fine-tuning.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

Bottou, L., Curtis, F. E., and Nocedal, J. Optimization methods for large-scale machine learning. *SIAM review*, 60(2):223–311, 2018.

Cheng, G., Han, J., and Lu, X. Remote sensing image scene classification: Benchmark and state of the art. *Proceedings of the IEEE*, 105(10):1865–1883, 2017.

Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., and Vedaldi, A. Describing textures in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 3606–3613, 2014.

Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Dettmers, T., Lewis, M., Belkada, Y., and Zettlemoyer, L. Gpt3. int8 (): 8-bit matrix multiplication for transformers at scale. *Advances in neural information processing systems*, 35:30318–30332, 2022.

Dettmers, T., Svirschevski, R. A., Egiazarian, V., Kuznedelev, D., Frantar, E., Ashkboos, S., Borzunov, A., Hoefler, T., and Alistarh, D.-A. Spqr: A sparse-quantized representation for near-lossless llm weight compression. In *12th International Conference on Learning Representations*, 2024.

Ding, N., Qin, Y., Yang, G., Wei, F., Yang, Z., Su, Y., Hu, S., Chen, Y., Chan, C.-M., Chen, W., et al. Parameter-efficient fine-tuning of large-scale pre-trained language models. *Nature machine intelligence*, 5(3):220–235, 2023.

Dodge, J., Ilharco, G., Schwartz, R., Farhadi, A., Hajishirzi, H., and Smith, N. Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping. *arXiv preprint arXiv:2002.06305*, 2020.

Gast, R., Solla, S. A., and Kennedy, A. Neural heterogeneity controls computations in spiking neural networks. *Proceedings of the National Academy of Sciences*, 121(3): e2311885121, 2024.

Gerstner, W., Kistler, W. M., Naud, R., and Paninski, L. *Neuronal dynamics: From single neurons to networks and models of cognition*. Cambridge University Press, 2014.

Ghosh-Dastidar, S. and Adeli, H. Spiking neural networks. *International journal of neural systems*, 19(04):295–308, 2009.

Glorot, X. and Bengio, Y. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics*, pp. 249–256. JMLR Workshop and Conference Proceedings, 2010.

Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., Bi, X., et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Han, Z., Gao, C., Liu, J., Zhang, J., and Zhang, S. Q. Parameter-efficient fine-tuning for large models: A comprehensive survey. *arXiv preprint arXiv:2403.14608*, 2024.

Hayou, S., Ghosh, N., and Yu, B. Lora+: Efficient low rank adaptation of large models. *arXiv preprint arXiv:2402.12354*, 2024.

He, B., Noci, L., Paliotta, D., Schlag, I., and Hofmann, T. Understanding and minimising outlier features in transformer training. *Advances in Neural Information Processing Systems*, 37:83786–83846, 2024.

He, H., Ye, P., Ren, Y., Yuan, Y., Zhou, L., Ju, S., and Chen, L. Gora: Gradient-driven adaptive low rank adaptation. *arXiv preprint arXiv:2502.12171*, 2025.

Helber, P., Bischke, B., Dengel, A., and Borth, D. Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7):2217–2226, 2019.

Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., and Steinhardt, J. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021.

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., Casas, D. d. L., Hendricks, L. A., Welbl, J., Clark, A., et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

Houben, S., Stallkamp, J., Salmen, J., Schlipsing, M., and Igel, C. Detection of traffic signs in real-world images: The german traffic sign detection benchmark. In *The 2013 international joint conference on neural networks (IJCNN)*, pp. 1–8. Ieee, 2013.

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.

Hu, Z., Wang, L., Lan, Y., Xu, W., Lim, E.-P., Bing, L., Xu, X., Poria, S., and Lee, R. Llm-adapters: An adapter family for parameter-efficient fine-tuning of large language models. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, pp. 5254–5276, 2023.

Huang, X. S., Perez, F., Ba, J., and Volkovs, M. Improving transformer optimization through better initialization. In *International Conference on Machine Learning*, pp. 4475–4483. PMLR, 2020.

Jiang, A. Q. et al. Mistral 7b, arxiv. org, 2023. *Search in*.

Jiang, S., Liao, Y., Zhang, Y., Wang, Y., and Wang, Y. Fine-tuning with reserved majority for noise reduction. In *The Thirteenth International Conference on Learning Representations*, 2025.

Krause, J., Stark, M., Deng, J., and Fei-Fei, L. 3d object representations for fine-grained categorization. In *Proceedings of the IEEE international conference on computer vision workshops*, pp. 554–561, 2013.

Kumar, A., Raghunathan, A., Jones, R., Ma, T., and Liang, P. Fine-tuning can distort pretrained features and underperform out-of-distribution. *arXiv preprint arXiv:2202.10054*, 2022.

Lee, C., Jin, J., Kim, T., Kim, H., and Park, E. Owq: Outlier-aware weight quantization for efficient fine-tuning and inference of large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 13355–13364, 2024.

Li, S., Luo, X., Wang, H., Tang, X., Cui, Z., Liu, D., Li, Y., He, X., and Li, R. Beyond higher rank: Token-wise input-output projections for efficient low-rank adaptation. *arXiv preprint arXiv:2510.23123*, 2025.

Liao, Z., Liu, Y., Zheng, Q., and Pan, G. Spiking nerf: Representing the real-world geometry by a discontinuous representation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 13790–13798, 2024.

Lin, J., Tang, J., Tang, H., Yang, S., Chen, W.-M., Wang, W.-C., Xiao, G., Dang, X., Gan, C., and Han, S. Awq: Activation-aware weight quantization for on-device llm compression and acceleration. *Proceedings of machine learning and systems*, 6:87–100, 2024.

Liu, S.-Y., Wang, C.-Y., Yin, H., Molchanov, P., Wang, Y.-C. F., Cheng, K.-T., and Chen, M.-H. Dora: Weight-decomposed low-rank adaptation. In *Forty-first International Conference on Machine Learning*, 2024.

Lubana, E. S., Dick, R., and Tanaka, H. Beyond batchnorm: Towards a unified understanding of normalization in deep learning. *Advances in Neural Information Processing Systems*, 34:4778–4791, 2021.

Meng, F., Wang, Z., and Zhang, M. Pissa: Principal singular values and singular vectors adaptation of large language models. *Advances in Neural Information Processing Systems*, 37:121038–121072, 2024.

Mosbach, M., Andriushchenko, M., and Klakow, D. On the stability of fine-tuning bert: Misconceptions, explanations, and strong baselines. *arXiv preprint arXiv:2006.04884*, 2020.

Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. Reading digits in natural images with unsupervised feature learning. In *NIPS workshop on deep learning and unsupervised feature learning*, volume 2011, pp. 7. Granada, 2011.

Pennington, J. and Worah, P. Nonlinear random matrix theory for deep learning. *Advances in neural information processing systems*, 30, 2017.

Ponkshe, K., Singhal, R., Gorbunov, E., Tumanov, A., Horvath, S., and Vepakomma, P. Initialization using update approximation is a silver bullet for extremely efficient low-rank fine-tuning. *arXiv preprint arXiv:2411.19557*, 2024.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PmLR, 2021.

Shen, S., Yao, Z., Gholami, A., Mahoney, M., and Keutzer, K. Powernorm: Rethinking batch normalization in transformers. In *International conference on machine learning*, pp. 8741–8751. PMLR, 2020.

Team, G., Riviere, M., Pathak, S., Sessa, P. G., Hardin, C., Bhupatiraju, S., Hussenot, L., Mesnard, T., Shahriari, B., Ramé, A., et al. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.

Valipour, M., Rezagholizadeh, M., Kobyzev, I., and Ghodsi, A. Dylora: Parameter-efficient tuning of pre-trained models using dynamic search-free low-rank adaptation. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pp. 3274–3287, 2023.

Wang, H., Ma, S., Dong, L., Huang, S., Zhang, D., and Wei, F. Deepnet: Scaling transformers to 1,000 layers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(10):6761–6774, 2024a.

Wang, S., Yu, L., and Li, J. Lora-ga: Low-rank adaptation with gradient approximation. *Advances in Neural Information Processing Systems*, 37:54905–54931, 2024b.

Xiao, G., Lin, J., Seznec, M., Wu, H., Demouth, J., and Han, S. Smoothquant: Accurate and efficient post-training quantization for large language models. In *International conference on machine learning*, pp. 38087–38099. PMLR, 2023.

Xiao, J., Hays, J., Ehinger, K. A., Oliva, A., and Torralba, A. Sun database: Large-scale scene recognition from abbey to zoo. In *2010 IEEE computer society conference on computer vision and pattern recognition*, pp. 3485–3492. IEEE, 2010.

Xu, L., Xie, H., Qin, S.-Z. J., Tao, X., and Wang, F. L. Parameter-efficient fine-tuning methods for pretrained language models: A critical review and assessment. *arXiv preprint arXiv:2312.12148*, 2023.

Yan, J., Liu, Q., Zhang, M., Feng, L., Ma, D., Li, H., and Pan, G. Efficient spiking neural network design via neural architecture search. *Neural Networks*, 173:106172, 2024.

Yao, Z., Yazdani Aminabadi, R., Zhang, M., Wu, X., Li, C., and He, Y. Zeroquant: Efficient and affordable post-training quantization for large-scale transformers. *Advances in neural information processing systems*, 35: 27168–27183, 2022.

You, K., Liu, Y., Zhang, Z., Wang, J., Jordan, M. I., and Long, M. Ranking and tuning pre-trained models: A new paradigm for exploiting model hubs. *Journal of Machine Learning Research*, 23(209):1–47, 2022.

Yu, L., Jiang, W., Shi, H., Yu, J., Liu, Z., Zhang, Y., Kwok, J. T., Li, Z., Weller, A., and Liu, W. Metamath: Bootstrap your own mathematical questions for large language models. *arXiv preprint arXiv:2309.12284*, 2023.

Yuan, Z., Shang, Y., and Dong, Z. Pb-llm: Partially binarized large language models. In *The Twelfth International Conference on Learning Representations*.

Zhang, B. and Sennrich, R. Root mean square layer normalization. *Advances in neural information processing systems*, 32, 2019.

Zhang, Q., Chen, M., Bukharin, A., Karampatziakis, N., He, P., Cheng, Y., Chen, W., and Zhao, T. Adalora: Adaptive budget allocation for parameter-efficient fine-tuning. *arXiv preprint arXiv:2303.10512*, 2023.

Zhang, W., Li, Z., Ma, D., Tang, H., Jiang, X., Zheng, Q., and Pan, G. Spiking gs: Towards high-accuracy and low-cost surface reconstruction via spiking neuron-based gaussian splatting. *arXiv preprint arXiv:2410.07266*, 2024.

Zheng, H., Zheng, Z., Hu, R., Xiao, B., Wu, Y., Yu, F., Liu, X., Li, G., and Deng, L. Temporal dendritic heterogeneity incorporated with spiking neural networks for learning multi-timescale dynamics. *Nature Communications*, 15 (1):277, 2024.

A. Proof of Proof of Heuristic Corollary 3.2

A.1. Empirical Observations and Modeling

1. Let the feature matrix be $\mathbf{Z}_P^l \in \mathbb{R}^{n\times d}$, with covariance matrix $\Sigma^l = \mathrm{Cov}(\mathbf{Z}_P^l)$.
2. **Empirical characterization.** Empirical analysis of LlaMA2/3 and Mistral-7B reveals that the ℓ_2-aggregated energy of singular values concentrates on a low-dimensional subspace S with $|S| = s \approx 3$–10 and concentration coefficient $\kappa \in [1.5, 2.5]$. This observed LLM behavior is characterized as:
$$\sqrt{\sum_{i\in S}\sigma_i^2} \;\geq\; \kappa\sqrt{\sum_{i\notin S}\sigma_i^2}, \qquad |S| = s \ll d,\ \kappa > 1,$$
where σ_i are the singular values of Σ^l. This formulation serves as an empirical characterization of LLM activation distributions, replacing the stronger condition of single singular value dominance ($\sigma_1 \geq \kappa \sum_{i=2}^d \sigma_i$).
3. The effective rank is defined as:
$$R_{\text{eff}}(\mathbf{Z}_P^l) = \frac{(\sum_{i=1}^d \sigma_i)^2}{\sum_{i=1}^d \sigma_i^2}.$$

A.2. Proof Procedure

The following proof first considers the basic case where the ℓ_2-aggregated energy concentrates on a single dimension ($s = 1$). This serves as a building block for the general subspace case ($s \approx 3$–10) presented in the main text. For mathematical tractability, we analyze this basic case using the condition $\sigma_1 \geq \kappa \sum_{i=2}^d \sigma_i$, which corresponds to the special case $s = 1$ of the empirical subspace dominance characterization.

Step 1: Establish Singular Value Relation From the anomalous weight characterization:
$$\sigma_1 \geq \kappa \sum_{i=2}^d \sigma_i.$$
Let $S = \sum_{i=2}^d \sigma_i$, then:
$$\sigma_1 \geq \kappa S.$$

Step 2: Compute the Numerator The total sum of singular values is:
$$\sum_{i=1}^d \sigma_i = \sigma_1 + S \geq \kappa S + S = (\kappa + 1)S.$$

Step 3: Compute the Denominator and Establish a Lower Bound Consider the lower bound of the sum of squares:
$$\sum_{i=1}^d \sigma_i^2 = \sigma_1^2 + \sum_{i=2}^d \sigma_i^2.$$
By the Cauchy–Schwarz inequality:
$$S^2 \leq (d-1)\sum_{i=2}^d \sigma_i^2.$$
Thus:
$$\sum_{i=2}^d \sigma_i^2 \geq \frac{S^2}{d-1}.$$
Substituting back:
$$\sum_{i=1}^d \sigma_i^2 \geq \sigma_1^2 + \frac{S^2}{d-1} \geq \kappa^2 S^2 + \frac{S^2}{d-1} = S^2\left(\kappa^2 + \frac{1}{d-1}\right).$$

Step 4: Derive Upper Bound for Effective Rank Combine the numerator and denominator:

$$R_{\text{eff}} = \frac{(\sum_{i=1}^{d} \sigma_i)^2}{\sum_{i=1}^{d} \sigma_i^2} \leq \frac{[(\kappa+1)S]^2}{S^2\left(\kappa^2 + \frac{1}{d-1}\right)} = \frac{(\kappa+1)^2}{\kappa^2 + \frac{1}{d-1}}.$$

Step 5: Asymptotic Analysis As dimension $d \to \infty$:

$$\lim_{d\to\infty} R_{\text{eff}} \leq \frac{(\kappa+1)^2}{\kappa^2} = 1 + \frac{2}{\kappa} + \frac{1}{\kappa^2}.$$

Define the constant $C = 1 + \frac{2}{\kappa} + \frac{1}{\kappa^2}$, then:

$$R_{\text{eff}}(\mathbf{Z}_P^l) \leq C \ll d.$$

Conclusion We first analyze the basic case of a single dimension ($s = 1$). In this case, the effective rank is bounded above by a dimension-independent constant C, showing that anomalous activations can lead to rank collapse.

Extension to the General Case ($s \approx 3 - 10$) TThe single-dimension analysis ($s = 1$) can be naturally extended to a more general low-dimensional subspace setting.

Empirical evidence from LLaMA2/3 and Mistral-7B justifies this extension: the top-5 singular values collectively account for 45% to 65% of the total ℓ_2 energy, whereas the largest singular value alone contributes only 17% to 28%. This indicates that the spectral energy is distributed across a low-dimensional subspace of size $s \approx 3 - 10$, rather than concentrated in a single direction. The concentration coefficient κ is empirically observed to lie within $[1.5, 2.5]$.

Building upon the analytical framework established for $s = 1$, a qualitatively analogous conclusion holds under this subspace setting: the effective rank collapses to $O(s) \ll d$, meaning that the representation is confined to a low-dimensional subspace whose dimensionality is independent of the full ambient dimension d.

General Bound Under this subspace setting, the effective rank satisfies:

$$R_{\text{eff}} = \frac{(\sum \sigma_i)^2}{\sum \sigma_i^2} \;\leq\; \left(\sqrt{s} + \frac{\sqrt{r}}{\kappa}\right)^2, \quad r = d - s.$$

Derivation. Let $A = \sum_{i\in S} \sigma_i$, $B = \sum_{i\notin S} \sigma_i$, and $A_2 = \sum_{i\in S} \sigma_i^2$, $B_2 = \sum_{i\notin S} \sigma_i^2$. By Cauchy–Schwarz, $A \leq \sqrt{sA_2}$, $B \leq \sqrt{rB_2}$. From $\sqrt{A_2} \geq \kappa\sqrt{B_2}$ we have $\sqrt{B_2} \leq \sqrt{A_2}/\kappa$. Thus $A + B \leq \sqrt{A_2}(\sqrt{s} + \sqrt{r}/\kappa)$. Since $A_2 + B_2 \geq A_2$, we obtain the bound.

Example (LlaMA2-7B, $s = 5, \kappa = 2, d = 4096$**)**

$$R_{\text{eff}} \leq (\sqrt{5} + 32)^2 \approx 1172 \ll 4096.$$

Remark The above derivation provides an illustrative bound. Its purpose is to capture the qualitative direction of the effect and to provide intuition for the design of NeuroReg.

B. Proof of Heuristic Corollary 3.3

B.1. Definitions and Assumptions

1. Consider the gradient update of the weight increment $\Delta\mathbf{W}^l$. The loss function $\mathcal{L}$ satisfies standard Lipschitz continuity and smoothness assumptions, which is a standard assumption in optimization theory for analyzing gradient dynamics(Bottou et al., 2018) and is commonly adopted in fine-tuning studies(Hayou et al., 2024).

2. The forward pass generates an abnormal activation $\mathbf{z}_{P,\text{abnormal}}^l$ satisfying $\left\|\mathbf{z}_{P,\text{abnormal}}^l\right\| \geq \xi$.

3. The network structure adheres to the chain rule for gradient backpropagation, where the activation $\mathbf{z}_P^l$ is linked to the gradient $\mathbf{g}_P^l$ via the network's Jacobian matrix $\mathbf{J}^l$: $\mathbf{g}_P^l = (\mathbf{J}^l)^\top \frac{\partial \mathcal{L}}{\partial \mathbf{z}_P^l}$.

B.2. Proof Procedure

Step 1: Expansion of the Gradient Expression Consider the gradient of $\Delta \mathbf{W}^l$:

$$\frac{\partial \mathcal{L}}{\partial \Delta \mathbf{W}^l} = \mathbf{g}_P^l (\mathbf{z}_P^{l-1})^\top$$

where $\mathbf{g}_P^l = \frac{\partial \mathcal{L}}{\partial \mathbf{z}_P^l}$ is the backpropagated gradient at layer l.

Step 2: Dominant Influence of Abnormal Activation According to the definition of abnormal activation, $\mathbf{z}_P^l$ can be decomposed as:

$$\mathbf{z}_P^l = \mathbf{z}_{P,\text{normal}}^l + \mathbf{z}_{P,\text{abnormal}}^l$$

with $\|\mathbf{z}_{P,\text{abnormal}}^l\| \geq \xi$ and $\|\mathbf{z}_{P,\text{abnormal}}^l\| \gg \|\mathbf{z}_{P,\text{normal}}^l\|$.

Since $\mathbf{z}_{P,\text{abnormal}}^l$ dominates $\mathbf{z}_P^l$, the gradient of the loss with respect to $\mathbf{z}_P^l$, $\frac{\partial \mathcal{L}}{\partial \mathbf{z}_P^l}$, exhibits a significant component along the direction of $\mathbf{z}_{P,\text{abnormal}}^l$. Let $\mathbf{u}$ be the unit vector in the direction of $\mathbf{z}_{P,\text{abnormal}}^l$. Then:

$$\left\langle \frac{\partial \mathcal{L}}{\partial \mathbf{z}_P^l}, \mathbf{u} \right\rangle \geq \gamma \xi$$

where $\gamma > 0$ is a constant related to the curvature of the loss function.

Step 3: Projection onto the Anomalous Weight Direction Project the gradient onto the anomalous weight direction $\mathbf{W}_{\text{abnormal}}^l$:

$$\text{Proj}_{\mathbf{W}_{\text{abnormal}}^l}\left(\frac{\partial \mathcal{L}}{\partial \Delta \mathbf{W}^l}\right) = \mathbf{W}_{\text{abnormal}}^l (\mathbf{W}_{\text{abnormal}}^l)^\dagger \cdot \mathbf{g}_P^l (\mathbf{z}_P^{l-1})^\top$$

where $(\cdot)^\dagger$ denotes the pseudo-inverse.

Let $\mathbf{v}$ be the principal singular vector direction of $\mathbf{W}_{\text{abnormal}}^l$, and $\sigma_{\max}$ the corresponding maximum singular value. Since $\mathbf{z}_{P,\text{abnormal}}^l$ aligns with the direction of $\mathbf{W}_{\text{abnormal}}^l$ (by Definition 1), we have:

$$\mathbf{z}_{P,\text{abnormal}}^l \approx \mathbf{W}_{\text{abnormal}}^l \mathbf{z}_P^{l-1}$$

Thus, $\mathbf{z}_P^{l-1}$ also has a component aligned with $\mathbf{v}$ in the input space of $\mathbf{W}_{\text{abnormal}}^l$.

Step 4: Derivation of the Gradient Lower Bound Consider the Frobenius norm:

$$\left\| \text{Proj}_{\mathbf{W}_{\text{abnormal}}^l}\left(\frac{\partial \mathcal{L}}{\partial \Delta \mathbf{W}^l}\right) \right\|_F \geq \frac{1}{\sigma_{\max}} \left\| (\mathbf{W}_{\text{abnormal}}^l)^\top \mathbf{g}_P^l (\mathbf{z}_P^{l-1})^\top \right\|_F$$

Since the component of $\mathbf{g}_P^l$ in the direction of $\mathbf{z}_{P,\text{abnormal}}^l$ is at least $\gamma\xi$, and $\mathbf{z}_P^{l-1}$ aligns with $\mathbf{W}_{\text{abnormal}}^l$, we obtain:

$$\left\| (\mathbf{W}_{\text{abnormal}}^l)^\top \mathbf{g}_P^l (\mathbf{z}_P^{l-1})^\top \right\|_F \geq \gamma \xi \cdot \|\mathbf{z}_P^{l-1}\| \cdot \sigma_{\min}(\mathbf{W}_{\text{abnormal}}^l)$$

where $\sigma_{\min}$ is the minimum singular value.

Step 5: Final Lower Bound Combining the results above, let:

$$\alpha = \frac{\gamma \cdot \sigma_{\min}(\mathbf{W}_{\text{abnormal}}^l)}{\sigma_{\max}} \cdot \mathbb{E}[\|\mathbf{z}_P^{l-1}\|]$$

$$\beta = \text{constant term (contribution from normal activations)}$$

Then:

$$\left\| \text{Proj}_{\mathbf{W}_{\text{abnormal}}^l}\left(\frac{\partial \mathcal{L}}{\partial \Delta \mathbf{W}^l}\right) \right\|_F \geq \alpha \cdot \xi - \beta$$

Conclusion This proof demonstrates that abnormal activations cause disproportionately large updates in the anomalous weight direction through the gradient update process, thereby further reinforcing the anomalous weights in subsequent training steps and forming a self-amplifying cycle.

Remark: The constants α and β depend on the specific task loss, network architecture, and statistical properties of the activations. However, the existence of the linear dependency $\alpha \cdot \xi$ guarantees that abnormal activations inevitably lead to anomalous amplification of gradients.

Note: The above derivation provides an illustrative bound intended to capture the qualitative direction of the effect.

C. Computation Cost Analysis of NeuroReg

NeuroReg is a lightweight and efficient plug-and-play method with low computational cost, making it well-suited for large-scale language models. Its key characteristics are as follows:

- **Parameter Overhead**: Introduces less than 0.5% additional parameters, with minimal impact on the overall model size.
- **Memory Overhead**: The extra memory required for dynamic anomaly sensing and element-wise operations is less than 1%, which is nearly negligible.
- **Computation Cost**:
  - **Training Phase**: Compared to standard PEFT methods, training time increases by approximately 3–5%, while significantly improving stability and performance.
  - **Inference Phase**: Only affects the forward pass, with inference latency increasing by less than 1%.
- **Scalability**: Computational and memory costs grow linearly with model size, making it suitable for large models with billions of parameters.

Summary: NeuroReg achieves an excellent balance between computational efficiency and performance improvement, offering a cost-effective solution for fine-tuning large-scale language models.

D. Experimental Settings

Evaluation on Language Understanding Tasks: Following the setup of Hu et al.(Hu et al., 2023), we combine the training sets from all eight tasks into a single training corpus, while evaluation is performed separately on the held-out test set of each individual task. LoRA modules are inserted into the key, value, query, attention-output, and all feed-forward weight matrices, using a rank of $r = 32$.

Evaluation on Mathematical Reasoning Tasks: Experiments employ a unified LoRA configuration, where adapters are applied to the query, key, value, and output weight matrices of attention modules, as well as to all feed-forward layers, with rank values $r = 32$.

Environment: All experiments were conducted on NVIDIA A100 80GB GPUs with CUDA 12.1 and PyTorch 2.1.0.

E. Experimental Results

E.1. Commonsense reasoning tasks on Llama3-8B.

We fine-tune LlaMA3-8B on a collection of commonsense reasoning tasks comprising eight sub-tasks, each with its own predefined training and test splits. The results are summarized in Table A.1.

NeuroReg consistently improves the performance of all PEFT methods it is combined with. When integrated with standard LoRA, NeuroReg+LoRA achieves an average accuracy of 82.2%, outperforming vanilla LoRA (80.8%) by 1.4%. For the relatively weaker baseline PiSSA, NeuroReg+PiSSA achieves 78.0%, a substantial gain of 2.6% over PiSSA (75.4%), demonstrating NeuroReg's effectiveness in stabilizing training for methods more susceptible to anomalous activations.

Table A.1. Performance comparison of LlaMA3-8B with various PEFT methods on eight commonsense reasoning datasets.

PEFT Method	BoolQ	PIQA	SIQA	HellaSwag	WinoGrande	ARC-e	ARC-c	OBQA	Avg.
LoRA	70.8	85.2	79.9	91.7	84.3	84.2	71.2	79.0	80.8
PiSSA	67.1	81.1	77.2	83.6	78.9	77.7	63.2	74.6	75.4
DoRA	74.6	89.3	79.9	95.5	85.6	90.5	80.4	85.8	85.2
TopLoRA	74.3	90.1	83.4	94.5	88.0	82.0	94.0	88.4	86.9
NeuroReg+LoRA	72.0	86.2	81.0	93.5	85.5	85.8	73.0	80.5	82.2
NeuroReg+PiSSA	69.5	83.0	79.0	87.0	81.0	80.0	66.5	77.5	78.0
NeuroReg+DoRA	**75.5**	90.0	81.0	**96.3**	86.5	**91.0**	81.0	87.5	86.1
NeuroReg+TopLoRA	75.0	**90.8**	**84.2**	95.5	**88.8**	82.7	**94.8**	**89.0**	**87.6**

Notably, NeuroReg also delivers consistent improvements over strong baselines and on challenging benchmarks. NeuroReg+DoRA attains an average accuracy of 86.1%, slightly outperforming standalone DoRA (85.2%) by 0.9%, with particularly notable improvements on HellaSwag (96.3% vs. 95.5%) and OBQA (87.5% vs. 85.8%). Similarly, NeuroReg+TopLoRA achieves a new state-of-the-art average of 87.6%, surpassing the already strong TopLoRA baseline (86.9%) by 0.7%, while maintaining competitive performance on ARC-c (94.8%) and HellaSwag (95.5%). The results further highlight NeuroReg's plug-and-play nature: it can be seamlessly integrated with diverse PEFT strategies—including standard LoRA, PiSSA, DoRA, and TopLoRA—without requiring method-specific adjustments, delivering consistent performance gains across the board.

Compared with the results on LLaMA2-7B, LLaMA3-8B exhibits more pronounced performance gains when enhanced with NeuroReg. For instance, NeuroReg+LoRA yields a 1.4% improvement on LLaMA3-8B, compared to only 0.6% on LLaMA2-7B; NeuroReg+TopLoRA achieves a 0.7% gain on LLaMA3-8B, versus 0.3% on LLaMA2-7B. This observation validates that despite more extensive pre-training, the increased parameter count in larger models (8B vs. 7B) introduces more complex anomalous activation distributions, allowing NeuroReg's dynamic outlier suppression mechanism to exert a greater effect.

E.2. Evaluation on Image Classification Tasks

We evaluate the performance of NeuroReg-enhanced PEFT methods on eight image classification benchmarks by comparing our approach with several state-of-the-art PEFT baselines. Experiments are conducted by fine-tuning the CLIP-ViT-B/16 visual backbone (Radford et al., 2021) on datasets including StanfordCars (Krause et al., 2013), DTD (Cimpoi et al., 2014), EuroSAT (Helber et al., 2019), GTSRB (Houben et al., 2013), RESISC45 (Cheng et al., 2017), SUN397 (Xiao et al., 2010), and SVHN (Netzer et al., 2011). The text classifier remains frozen using prompt templates such as "a photo of a {class}." Accuracy serves as the evaluation metric.

The results are summarized in Table A.2. NeuroReg consistently improves the base PEFT methods across almost all datasets. Notably, NeuroReg+LoRA-GA achieves the highest average accuracy of 88.9%, surpassing the strong baseline LoRA-GA (88.6%) by 0.3% and outperforming standard LoRA (83.4%) by 5.5%. Moreover, NeuroReg+LoRA achieves 83.9%, exceeding vanilla LoRA (83.4%) by 0.5% and slightly outperforming LoRA+ (83.5%) and DoRA (83.6%). These gains are observed even in challenging fine-grained recognition tasks such as StanfordCars and RESISC45, where NeuroReg+LoRA-GA reaches 85.5% and 94.2%, respectively. The consistent improvements across diverse architectures, including LoRA, DoRA, and LoRA-GA, demonstrate that NeuroReg serves as a lightweight, plug-in regularizer that enhances training stability and generalization without introducing substantial computational overhead.

F. Supplementary Results for Training Dynamics Analysis

Due to space constraints, we present the comparative results of training dynamics, including the evolution of rank, layer-wise gradient norm distributions, and loss curves, for LoRA-SB vs. NeuroReg+LoRA-SB, TopLoRA vs. NeuroReg+TopLoRA, and Norm vs. NeuroReg+Norm in the figures A1-A7.

Table A.2. Performance of fine-tuning CLIP-VIP-B/16 on 7 image classification tasks.

PEFT Method	Cars	DTD	EuroSAT	GTSRB	RESISC45	SUN397	SVHN	Avg.
LoRA	72.8	73.9	96.9	92.4	90.0	70.1	88.0	83.4
LoRA+	72.9	74.1	97.0	92.4	90.0	70.2	88.1	83.5
DoRA	73.7	73.7	97.0	92.4	90.0	70.2	88.2	83.6
LoRA-GA	85.2	77.5	98.0	95.3	94.4	75.4	94.6	88.6
NeuroReg+LoRA	73.4	74.3	97.2	93.5	90.4	70.3	88.4	83.9
NeuroReg+LoRA+	72.5	74.3	97.6	92.6	82.7	90.3	70.5	82.9
NeuroReg+DoRA	73.4	73.9	97.4	92.2	90.5	70.6	88.8	83.8
NeuroReg+LoRA-GA	**85.5**	**77.9**	**98.3**	**95.7**	94.2	75.6	**95.0**	**88.9**

Table A.3. Performance comparison of PEFT methods across different models and datasets. All results are accuracy (%).

PEFT Method	BoolQ	PIQA	SIQA	HellaSwag	WinoGrande	ARC-e	ARC-c	OBQA	Avg.
NeuroReg+LoRA*	68.1	78.7	78.5	83.6	82.3	79.1	63.1	80.0	76.7
NeuroReg+LoRA$^+$	69.8	79.6	79.6	85.3	84.5	79.1	64.8	81.1	78.0
NeuroReg+LoRA$^\diamond$	68.8	78.4	79.8	84.1	82.8	79.3	64.4	81.0	77.8
NeuroReg+LoRA	70.2	80.4	79.8	85.4	83.2	80.2	65.2	81.4	78.2

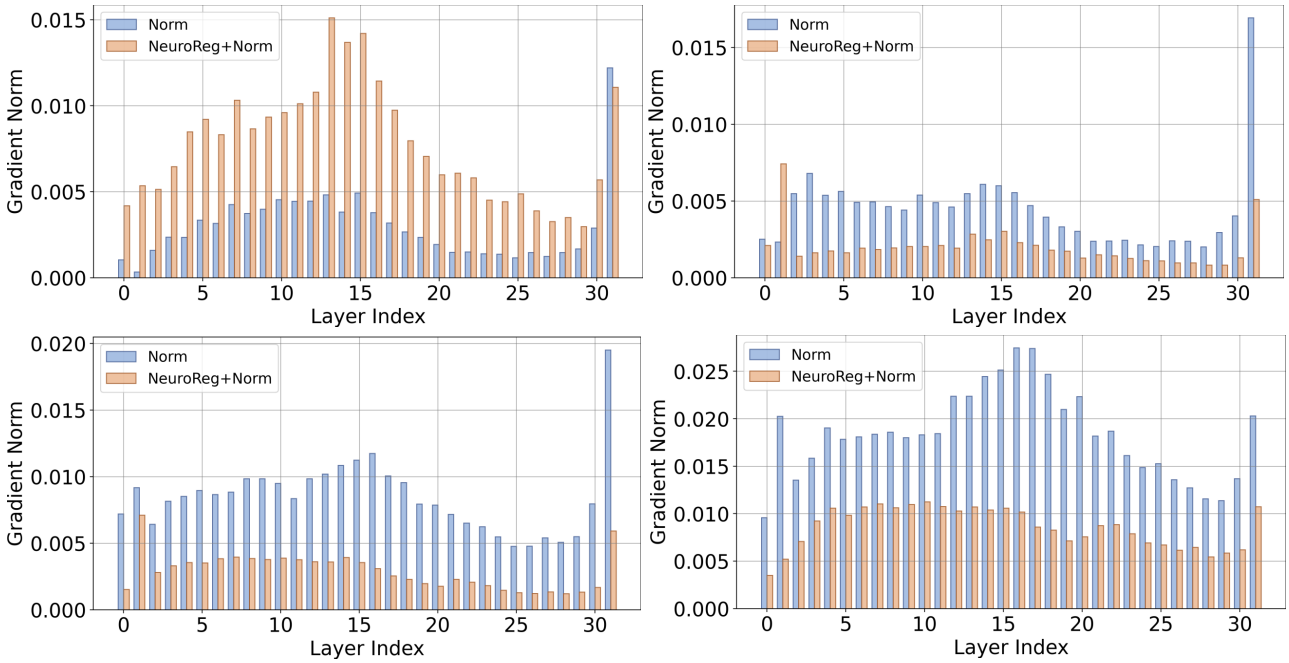

Figure A1. Comparison of gradient norms in different layers at training steps 3,200, 12,800, 25,600, and 51,200 for Norm and NeuroReg+Norm.

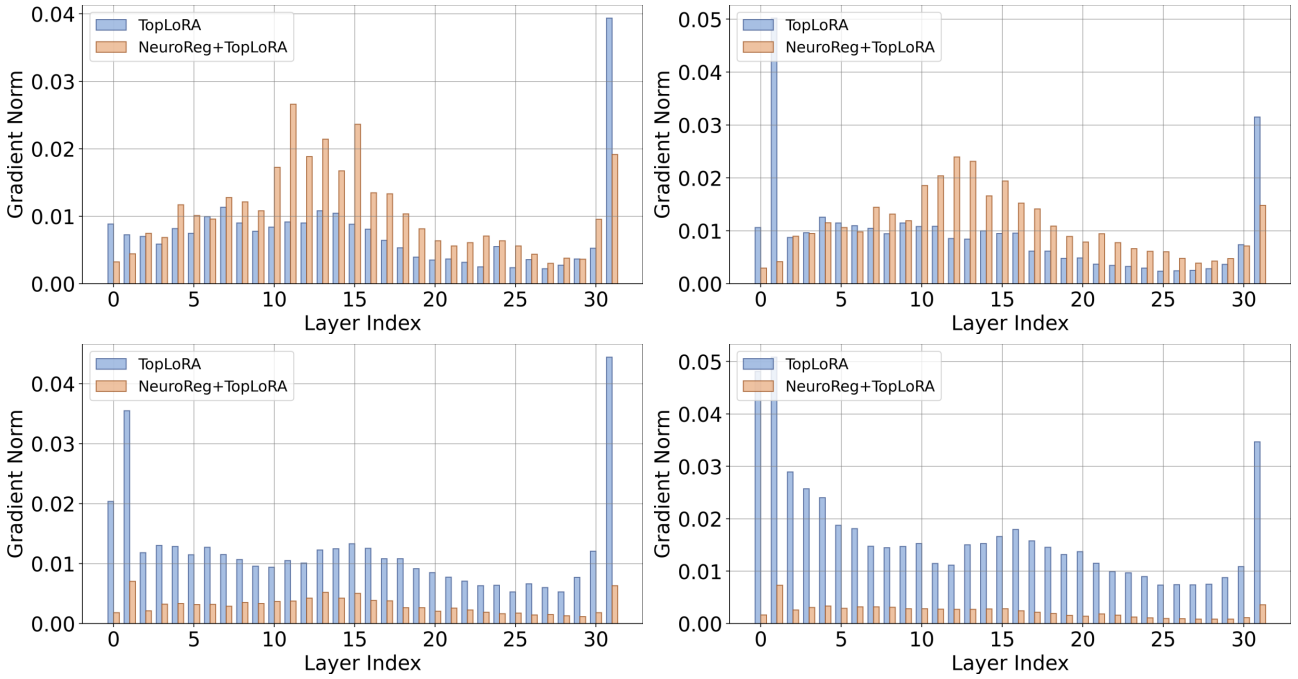

Figure A2. Comparison of gradient norms in different layers at training steps 3,200, 12,800, 25,600, and 51,200 for TopLoRA and NeuroReg+TopLoRA.

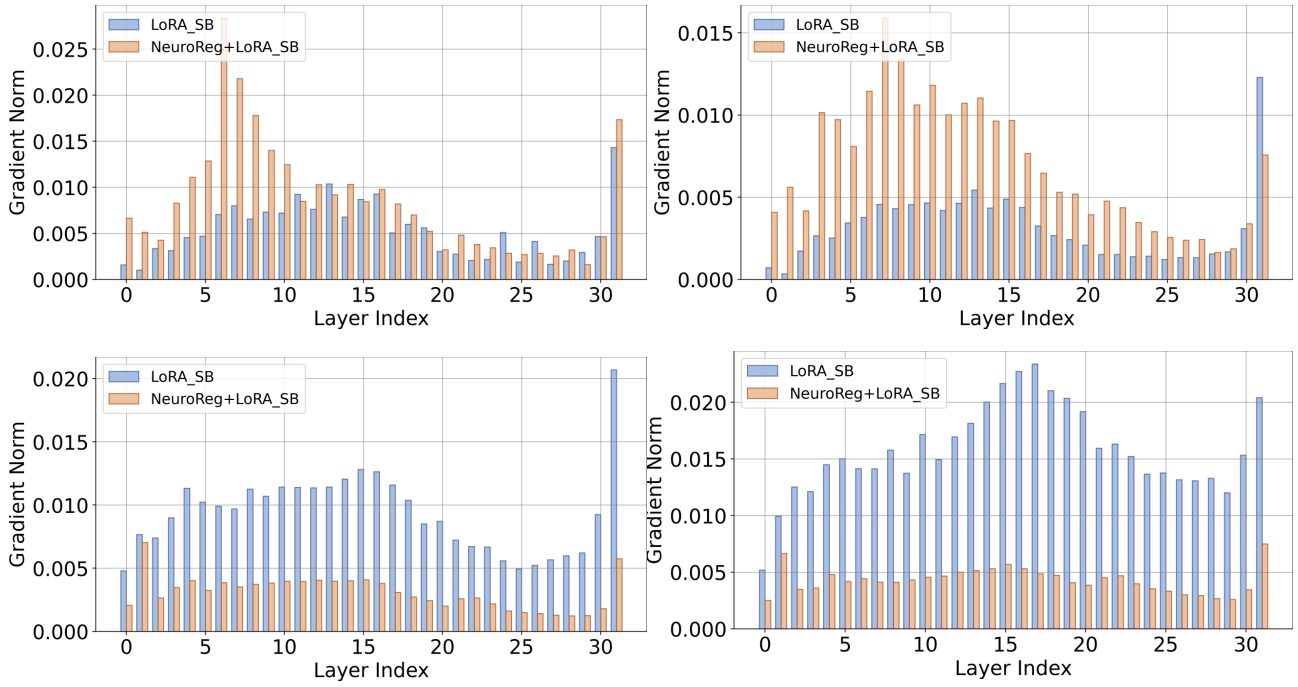

Figure A3. Comparison of gradient norms in different layers at training steps 3,200, 12,800, 25,600, and 51,200 for LoRA-SB and NeuroReg+LoRA-SB.

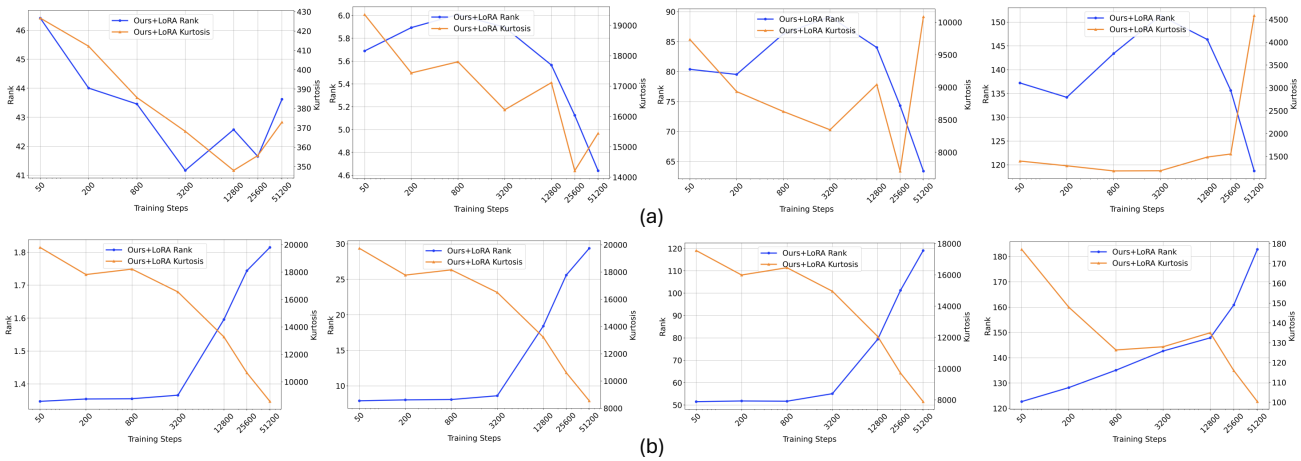

Figure A4. Evolution of activation kurtosis and effective rank in LlaMA2-7B layers 5, 14, 23, and 32 during fine-tuning with LoRA-SB (a) and NeuroReg+LoRA-SB (b).

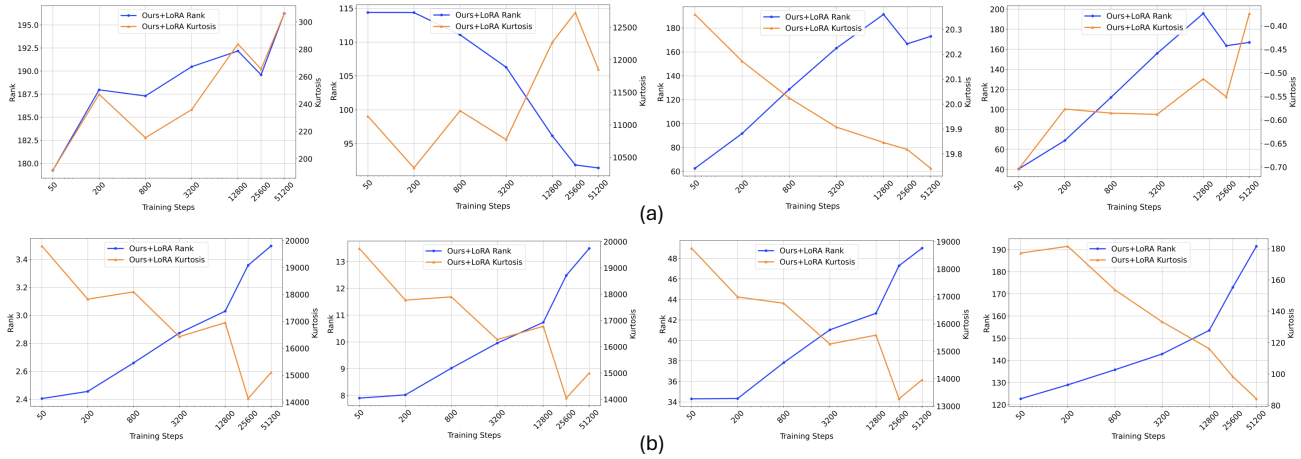

Figure A5. Evolution of activation kurtosis and effective rank in LlaMA2-7B layers 5, 14, 23, and 32 during fine-tuning with TopLoRA (a) and NeuroReg+TopLoRA (b).

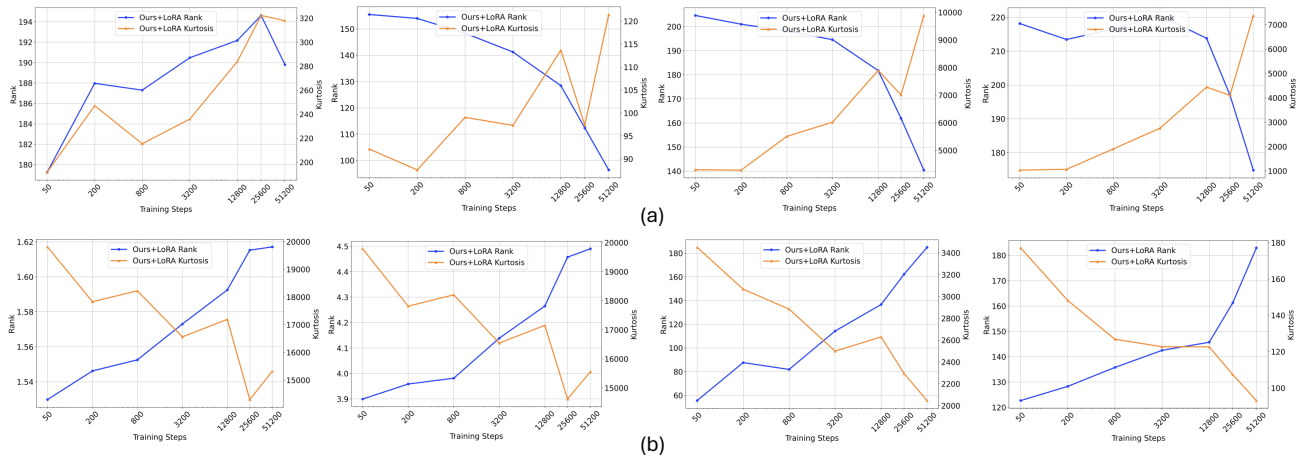

Figure A6. Evolution of activation kurtosis and effective rank in LlaMA2-7B layers 5, 14, 23, and 32 during fine-tuning with Norm (a) and NeuroReg+Norm (b).

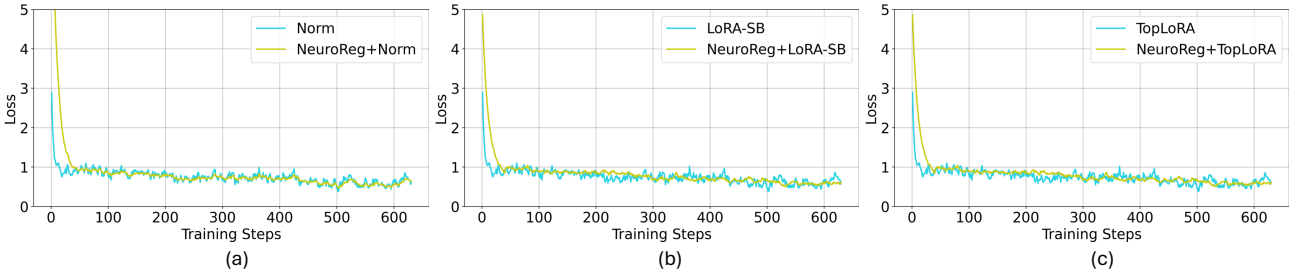

Figure A7. Training loss.